

Prove each limit using Cauchy's definition of limit.

1. $\lim_{x \rightarrow -3} (-7x - 5) = 16$ (This will give you some practice with some negative values but remember that ϵ and δ must be positive)

Proof: Let $\epsilon > 0$ be any positive real number.

Begin by assuming $|x+3| \leq 1$ so $-4 \leq x \leq -2$.

In this case $|f(x) - L| = |-7x - 5 - 16| = |-7x - 21| = |-7||x+3| = 7|x+3|$

The max value of 7 when $-4 \leq x \leq -2$ will be 7.

So if $|x+3| \leq 1$ we know $|f(x) - L| \leq 7|x+3| < \epsilon$ if $|x+3| < \frac{\epsilon}{7}$

Choose $\delta = \min \left\{ 1, \frac{\epsilon}{7} \right\}$

Therefore if $0 < |x+3| < \delta$ we have $|x+3| < 1$ and $|x+3| < \frac{\epsilon}{7}$

This implies $|f(x) - L| = 7|x+3| < 7 \cdot \frac{\epsilon}{7} = \epsilon$.

Thus $\lim_{x \rightarrow -3} (-7x - 5) = 16$

2. $\lim_{x \rightarrow 2} 3x^2 = 12$

Proof: Let $\epsilon > 0$ be any positive real number.

Begin by assuming $|x-2| \leq 1$ so $1 \leq x \leq 3$.

In this case $|f(x) - L| = |3x^2 - 12| = |3(x+2)(x-2)| = |3x+6||x-2|$

The max value of $|3x+6|$ when $1 \leq x \leq 3$ will be 15.

So if $|x-2| \leq 1$ we know $|f(x) - L| \leq 15|x-2| < \epsilon$ if $|x-2| < \frac{\epsilon}{15}$

Choose $\delta = \min \left\{ 1, \frac{\epsilon}{15} \right\}$.

Therefore if $0 < |x-2| < \delta$ we have $|x-2| < 1$ and $|x-2| < \frac{\epsilon}{15}$

This implies $|f(x) - L| = |3x+6||x-2| < 15 \cdot \frac{\epsilon}{15} = \epsilon$

Thus $\lim_{x \rightarrow 2} 3x^2 = 12$.

3. $\lim_{x \rightarrow 0} x^3 - 4x = 0$ (Be careful here when determining the max value of the other stuff. It does not occur at an endpoint.)

Let $\varepsilon > 0$ be any positive real number.

Begin by assuming $|x - 0| \leq 1$ so $-1 \leq x \leq 1$.

In this case $|f(x) - L| = |x^3 - 4x - 0| = |x^2 - 4| \cdot |x|$

The max value of $|x^2 - 4|$ when $-1 \leq x \leq 1$ will be 4

So if $|x| \leq 1$ we know $|f(x) - L| \leq 4|x| < \varepsilon$ if $|x| < \frac{\varepsilon}{4}$

Choose $\delta = \min \left\{ 1 \text{ and } \frac{\varepsilon}{4} \right\}$

Therefore if $0 < |x| < \delta$ we have $|x| < 1$ and $|x| < \frac{\varepsilon}{4}$

This implies $|f(x) - L| = |x^2 - 4||x| < 4 \cdot \frac{\varepsilon}{4} = \varepsilon$

Thus $\lim_{x \rightarrow 0} x^3 - 4x = 0$.